



PHY 412: STATISTICAL MECHANICS

First Course Handout: 25-26 Semester II

The responsibility to read and understand the First Course Handout lies entirely with the students.

1. Attendance:

- (a) 100% attendance is expected during the course. **Students with attendance less than 50% will be awarded an F grade.**
- (b) If a student misses a quiz or any other important event in the course, there will be no consideration.
- (c) If a student does not perform well, grace marks may be given, considering the student's attendance.

2. Grading:

- (a) Total marks (from all quizzes and exams) will be given out of 100.
- (b) Marks distribution: Assignments – 10%, Quizzes – 20%, Mid-sem – 30%, End-sem – 40%.
- (c) **A score of less than 30 (out of a total of 100) strictly corresponds to an F grade.**
- (d) The rest of the grades will be assigned relatively.

3. Homework Assignments:

- (a) Many practice problems will be given, and a subset of these will constitute the assignment.
- (b) Practice problems marked as assignments must be worked out and submitted by the due date.
- (c) Fully complete and timely submission is mandatory; any non-compliance will straightaway fetch the defaulter zero marks in the corresponding assignment problem.

4. Make-up/extra classes and Quizzes:

- (a) The instructor may miss a few classes that will be made up later in prescheduled makeup classes.
- (b) Quizzes will be conducted in prescheduled (extra) classes.

5. The following topics will be tentatively covered in the course:

Probability theory; Basic principles of Equilibrium Classical Statistical Mechanics, Micro-Canonical, Canonical and Grand Canonical ensembles; Random Walk, Brownian motion, Diffusion Equation, idea of Langevin and Fokker–Planck Equations; Ising model, Transfer Matrix method, Mean field theory; Quantum Statistical Mechanics, Density Matrix, Ideal Quantum Gases and their properties, Bose–Einstein Condensation, Free Electron gas; Phase Transitions, Curie-Weiss theory, Landau theory, Scaling near a critical point.

6. Reference Books:

- (a) Sethna - Statistical Mechanics: Entropy Order Parameters and Complexity
- (b) Kardar - Statistical Physics of Particles
- (c) Pathria, Beale - Statistical Mechanics
- (d) Huang - Statistical Mechanics